

# PatchCvT: Contextual Patch Convolutional Vision Transformer

## 1 Introduction

PatchCvT is a contextual patch-based representation architecture designed to estimate visual degradation at the object level by combining convolutional feature extraction, transformer-based contextual reasoning, and memory-bank nearest-neighbour comparison.

The central hypothesis is that individual convolutional feature vectors may be ambiguous when an object is degraded, partially occluded, rotated, affected by weather, or mixed with background information. Rather than evaluating each local feature independently, PatchCvT first allows the complete set of spatial CNN tokens to interact through a transformer encoder. The resulting contextual representations are then compared against a memory bank containing representative normal object representations.

The architecture therefore follows the sequence:

$$\boxed{\text{ROI} \rightarrow \text{ResNet18} \rightarrow 784 \text{ CNN tokens} \rightarrow \text{Transformer} \rightarrow \text{Memory Bank} \rightarrow \text{Nearest-Neighbour Anomaly Score}} \quad (1)$$

The anomaly score is used as a quantitative measure of representation degradation. It is not interpreted as an automatic rejection decision by the architecture itself.

## 2 Input Representation

The input is an object region of interest (ROI):

$$x \in \mathbb{R}^{H \times W \times 3} \quad (2)$$

where  $H$  and  $W$  denote the spatial dimensions of the ROI.

For the reference implementation, the ROI is resized to:

$$H = W = 224. \quad (3)$$

## 3 CNN Feature Extraction

A ResNet18 convolutional backbone extracts a spatial feature map:

$$F = f_{\theta}(x) \quad (4)$$

where the feature map used by PatchCvT is:

$$F \in \mathbb{R}^{28 \times 28 \times C}. \quad (5)$$

The spatial feature map is flattened along its spatial dimensions into a sequence of CNN feature tokens:

$$Q = \{q_j\}_{j=1}^N \quad (6)$$

with:

$$q_j \in \mathbb{R}^C. \quad (7)$$

Because the spatial feature map is  $28 \times 28$ , the number of tokens is:

$$N = 28 \times 28 = 784. \quad (8)$$

Therefore:

$$Q \in \mathbb{R}^{784 \times C}. \quad (9)$$

The complete sequence of 784 tokens is retained and passed to the transformer; it is not collapsed to a single ROI-level vector before contextual processing.

## 4 Contextual Transformer Encoding

The 784 CNN tokens are projected, when necessary, to the transformer embedding dimension  $D$ :

$$e_j = W_p q_j + b_p \quad (10)$$

where:

$$e_j \in \mathbb{R}^D. \quad (11)$$

The complete token sequence is:

$$E = [e_1, e_2, \dots, e_{784}] \in \mathbb{R}^{784 \times D}. \quad (12)$$

A positional encoding is added to preserve the spatial correspondence of the CNN tokens:

$$Z^{(0)} = E + P \quad (13)$$

where:

$$P \in \mathbb{R}^{784 \times D}. \quad (14)$$

The sequence is then processed by a transformer encoder:

$$Z = \text{Transformer}_\phi(Z^{(0)}) \quad (15)$$

with:

$$Z = \{z_j\}_{j=1}^{784}, \quad z_j \in \mathbb{R}^D. \quad (16)$$

The transformer performs contextual reasoning over all 784 spatial tokens. Each output token can therefore incorporate information from other spatial regions of the ROI.

## 5 Memory Bank Construction

The memory bank is constructed from contextual transformer representations extracted from training examples representing the reference or normal appearance of the target object.

Let the contextual token representations collected from the training data be:

$$\mathcal{Z}_{\text{train}} = \{z_j^{(i)}\} \quad (17)$$

where  $i$  indexes training ROIs and  $j$  indexes the 784 spatial tokens within each ROI.

A memory bank containing representative contextual feature vectors is defined as:

$$\mathcal{M} = \{m_k\}_{k=1}^K \quad (18)$$

where:

$$m_k \in \mathbb{R}^D. \quad (19)$$

The memory bank may contain a representative subset of the contextual training features rather than every training token. Its size  $K$  is therefore independent of the fixed number of 784 tokens produced for each ROI.

## 6 Nearest-Neighbour Memory Retrieval

At inference time, each contextual token  $z_j$  is compared against every memory vector  $m_k$ .

The squared Euclidean distance is:

$$d_{jk} = \|z_j - m_k\|_2^2. \quad (20)$$

For each token, the nearest memory representation is selected:

$$k_j^* = \arg \min_k d_{jk}. \quad (21)$$

The nearest-neighbour distance for token  $j$  is therefore:

$$s_j = d_{jk_j^*} = \min_k \|z_j - m_k\|_2^2. \quad (22)$$

The quantity  $s_j$  is the patch-level anomaly or degradation score.

A small value indicates that the contextual representation is close to a representation contained in the learned reference memory bank. A large value indicates that the representation deviates from the learned reference distribution.

## 7 ROI-Level Anomaly Score

Because each ROI produces 784 contextual token scores, a single ROI-level degradation score can be obtained by aggregating the token-level scores.

The primary aggregation is:

$$S_{\text{ROI}} = \max_{j \in \{1, \dots, 784\}} s_j. \quad (23)$$

Alternative aggregation functions may be evaluated experimentally, including the mean:

$$S_{\text{mean}} = \frac{1}{784} \sum_{j=1}^{784} s_j \quad (24)$$

and top- $k$  aggregation:

$$S_{\text{top}k} = \frac{1}{k} \sum_{j \in \text{Top}K(s)} s_j. \quad (25)$$

These alternatives are evaluation strategies rather than additional learned components of the architecture.

## 8 Interpretation of the Anomaly Score

The PatchCvT output is a continuous degradation score:

$$S_{\text{ROI}} \in [0, \infty). \quad (26)$$

The score is not constrained to  $[0, 1]$  and is not a probability.

Its interpretation is relative to the feature-space geometry induced by the trained CNN, transformer, and memory bank:

- low score: the ROI representation is close to learned reference representations;
- high score: the ROI representation deviates from the learned reference representations.

For experimental benchmarking, a binary decision may optionally be produced by applying a manually selected threshold:

$$\hat{y} = \begin{cases} 1, & S_{\text{ROI}} \geq T, \\ 0, & S_{\text{ROI}} < T, \end{cases} \quad (27)$$

where  $T$  is an evaluation threshold.

This threshold is not part of the learned PatchCvT architecture. It is introduced only for quantitative evaluation of whether the continuous degradation score agrees with manually assigned ground-truth labels.

## 9 End-to-End Formulation

Given an ROI  $x$ , the complete PatchCvT inference process is:

$$F = f_{\theta}(x) \quad (28)$$

$$Q = \text{Flatten}_{HW}(F) = \{q_j\}_{j=1}^{784} \quad (29)$$

$$E = \{W_p q_j + b_p\}_{j=1}^{784} \quad (30)$$

$$Z = \text{Transformer}_{\phi}(E + P) \quad (31)$$

$$s_j = \min_{m_k \in \mathcal{M}} \|z_j - m_k\|_2^2 \quad (32)$$

and finally:

$$S_{\text{ROI}} = \max_{j=1, \dots, 784} \min_{m_k \in \mathcal{M}} \|z_j - m_k\|_2^2 \quad (33)$$

Thus, the intended architecture is:

$$\boxed{\text{ResNet18} \rightarrow 784 \text{ CNN tokens} \rightarrow \text{Transformer} \rightarrow \text{Memory Bank} \rightarrow \text{Nearest-Neighbour Anomaly Score}} \quad (34)$$

## 10 Architectural Scope

PatchCvT intentionally does not contain a separate preservation head, preservation gate, robustness head, or ROI-level classification head.

The memory-bank distance is the fundamental anomaly/degradation signal.

The transformer is not used to directly classify the ROI as normal or anomalous. Its role is to transform the 784 local CNN representations into contextual representations before nearest-neighbour comparison with the learned memory bank.

The architecture therefore separates:

1. **Contextual representation learning:** ResNet18 extracts local spatial features and the transformer integrates contextual information across the 784 tokens.
2. **Memory-based degradation measurement:** Each contextual token is compared against the memory bank and its nearest-neighbour distance provides a continuous degradation signal.

This design preserves the distinction between representation learning and anomaly measurement. The anomaly score is produced by distance in contextual feature space, rather than by an additional learned classification head.